Weekly Business Review: Olist E-commerce

ELT pipeline, SQL analytics, data quality, and customer-satisfaction recommendation

Executive summary

The pipeline loaded all nine supplied CSVs and produced a one-row-per-order mart with **99,441 orders** and no duplicate order IDs. Item-price revenue across the observed data is **R$13,591,643.70** (freight excluded); mean available review score is **4.09/5**. The strongest descriptive customer-satisfaction signal is delivery lateness: **37.92%** of delivered orders rated 1 were late versus **3.00%** of those rated 5. This is an association, not proof of causation.

Pipeline and grain

Raw CSV tables are replaced on each run; staging and mart tables are rebuilt transactionally. Items, payments, and reviews are aggregated independently before joining to orders to avoid join fan-out. Blank timestamps become NULL; negative/missing item values are excluded from sums via explicit NULL handling. The mart preserves orders with no items/payments and records zero totals for absent item aggregates.

Quality-check results

SQL check	Severity	Failing rows	Disposition
customer match	CRITICAL	0	Passed; no action
delivery chronology	CRITICAL	0	Passed; no action
item numeric values	WARNING	0	Passed; no action
order id integrity	CRITICAL	0	Passed; no action
payment reconciliation	WARNING	1033	Flagged; investigate; not auto-corrected
review score	CRITICAL	0	Passed; no action

Payment reconciliation flagged 1,033 orders with payment total differing from item price + freight by more than R$0.05. This is a warning because vouchers, installments, and order/payment lifecycle differences may explain some discrepancies; inspect the saved affected-order samples before changing data. All configured critical checks passed on this run.

SQL findings and metric definitions

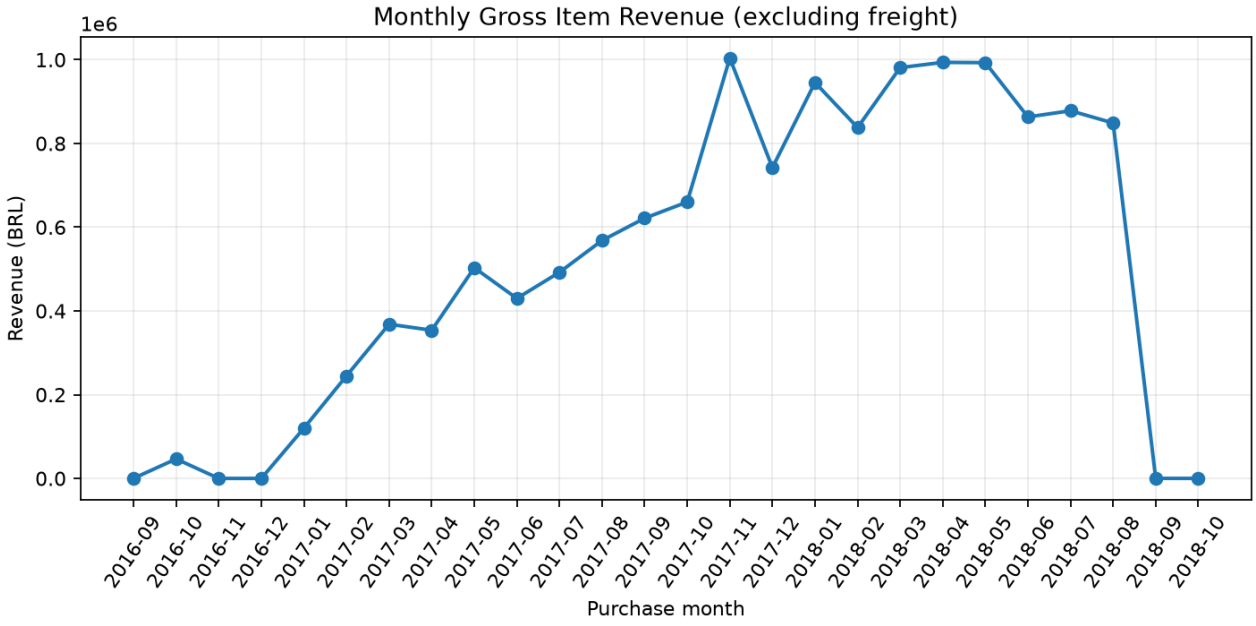
• **Monthly revenue:** monthly sum of order item prices (excluding freight), canceled orders excluded; LAG compares consecutive calendar months. Peak observed month: 2017-11 at R$1,003,862.14 (52.06% MoM where defined). September 2016 and October 2018 are partial edge months; interpret edge comparisons cautiously.

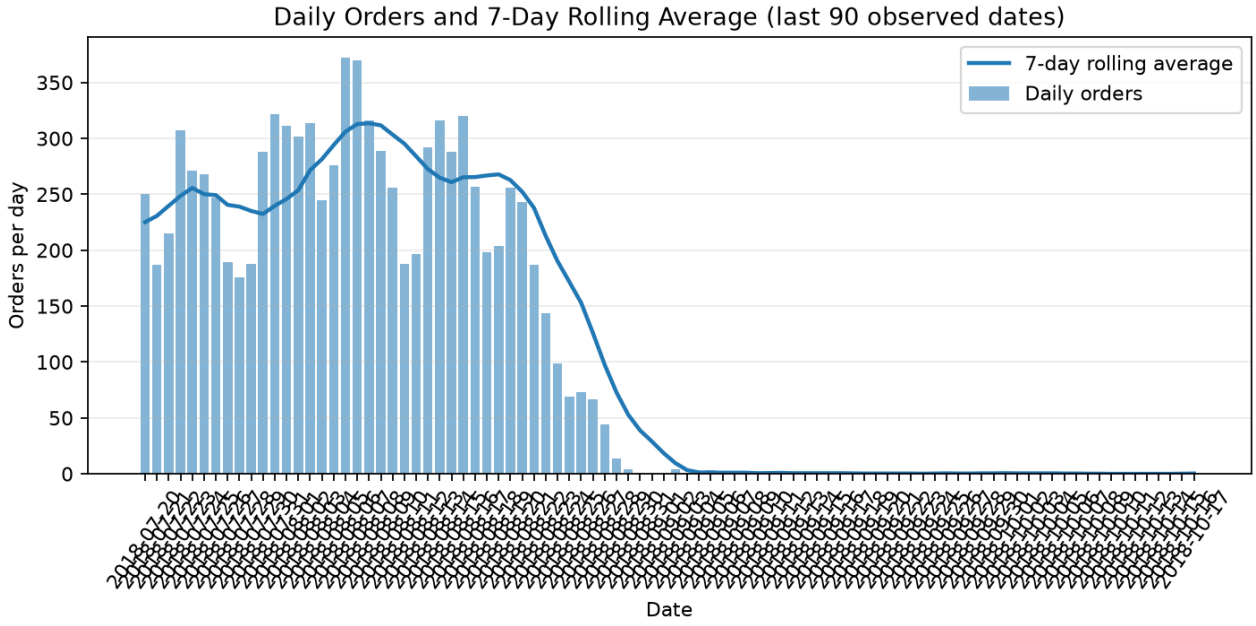
• **Repeat purchase rate:** 2,997 repeat customers / 96,096 identified customers = **3.12%**. Customer identity uses customer_unique_id; denominator includes identified customers with at least one order during the observed dataset window.

• **Category rankings:** DENSE_RANK within each customer state by category item-price revenue. In São Paulo (SP), the leading categories were bed_bath_table (R$477,794.20), health_beauty (rank 2), and watches_gifts (rank 3).

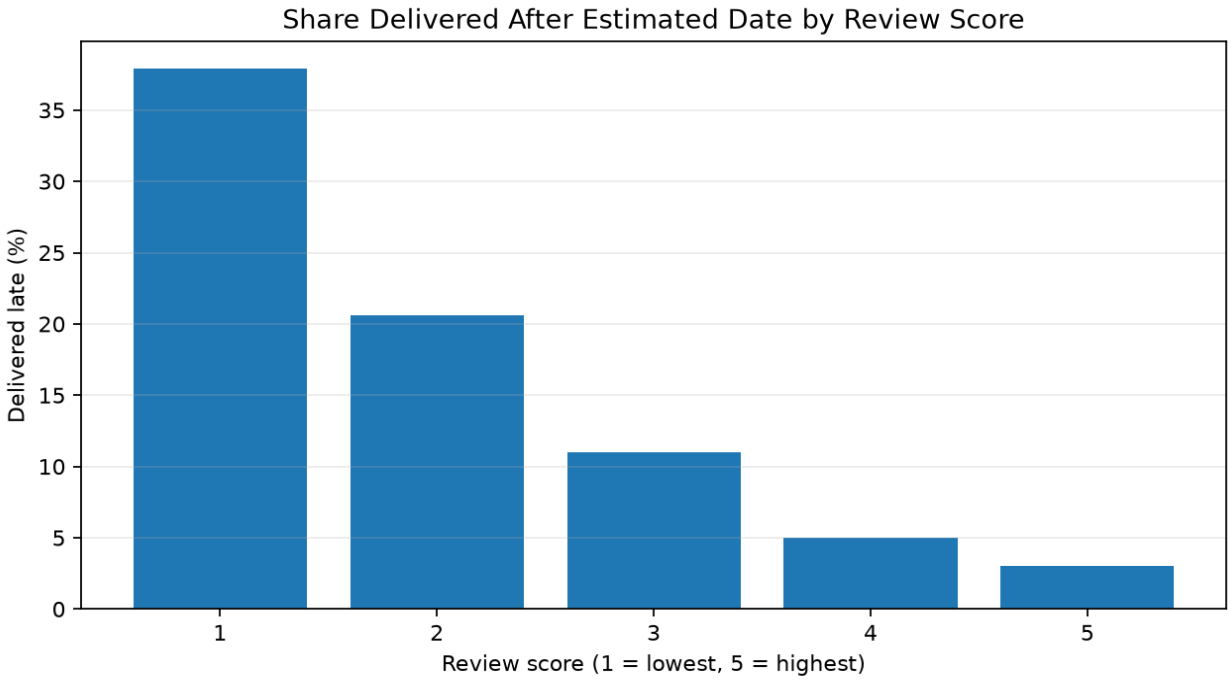
• **Rolling orders:** seven calendar days including zero-order dates; the first six dates have shorter windows, explicitly reported as days_in_window. **Delivery delay:** actual delivered date minus estimated delivery date; positive = late, negative = early.

Trend charts

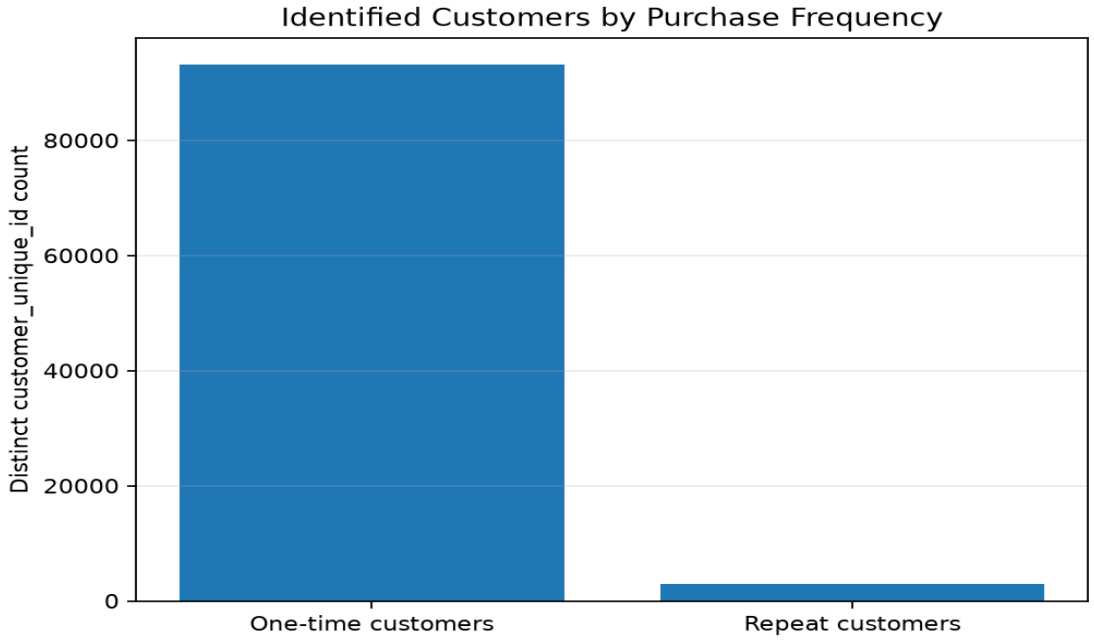




Customer and delivery experience



Orders rated 1 averaged -3.34 days relative to estimated delivery (i.e., delivered earlier on average, but closer to or later than the promise than five-star orders, which averaged -12.68 days). 34.92 percentage points separate the late-delivery rates for scores 1 and 5. This pattern is descriptive and may also reflect product, seller, or customer differences.



Metrics scorecard

Metric	Exact definition	Why it matters
North Star: successful delivered orders	Orders with order_status = delivered status/date and review_score ≥4; monthly count	Balances weekly demand with customer-perceived succe
On-time delivery rate	Delivered orders with actual delivery date ≤ estimated date / delivered orders with both dates	Operations guardrail for promise reliability.
Low-review rate	Delivered orders with review score ≤2 / delivered orders with a valid review score	Tracks customer dissatisfaction.
Repeat-purchase rate	Identified customer_unique_id with ≥2 distinct orders / identified customers	Signals retention, with a long observation window.
Payment reconciliation exception rate	Orders with |payment total − (item price + freight)| > R$0.05 / orders with payment records	Flags finance/checkout discrepancies without assuming

Implementation and limitations

Five Part C queries are saved separately and executed directly by SQLite; pandas is used only to export results and create charts. The source data covers 2016–2018, and monthly edge periods are incomplete. The review/delivery relationship is observational. Payment differences require business reconciliation rather than blind correction. The North Star definition is a proposed operational metric; no causal claim is made.

Memo to the Vice President

Subject: Reduce late deliveries to protect customer satisfaction

The Olist order history shows a clear association between delivery lateness and poor reviews. Among delivered orders with a valid review score and both actual and estimated delivery dates, **37.92% of one-star orders were delivered after the estimated date**, compared with **3.00% of five-star orders**—a 34.92 percentage-point difference. One-star orders averaged −3.34 days relative to the estimate, while five-star orders averaged −12.68 days. Negative values mean delivery before the estimate; the lower early-delivery buffer among low-rated orders suggests less margin for disruption. These figures are descriptive and do not establish that lateness alone causes low ratings.

Recommendation. Pilot a delivery-risk intervention for orders predicted or observed to be at risk of missing the estimated delivery date: trigger an operational alert to the seller/carrier, refresh the ETA, and proactively message the customer when the estimate changes. Start with a defined subset of sellers or lanes and compare with a matched or randomized holdout if operationally feasible.

Success KPI. Reduce the share of delivered orders arriving after the estimated date in the pilot cohort by at least 20% relative to baseline over an 8-week pilot, while monitoring the low-review rate (score ≤2) as a guardrail. Report weekly sample sizes and confidence intervals; do not claim success from small cohorts or a simple before/after change alone.

Risks and next steps. Validate data-quality exceptions and stratify results by seller, state, and order volume before scaling. The 1,033 payment reconciliation exceptions are not direct evidence of delivery problems and should be investigated separately. Historical Olist data is not current operational telemetry, so the pilot needs live delivery scans, ETA-change events, and consistent weekly measurement.